

# QuizAI: A Generative AI System for Automatic Multiple-Choice Quiz Generation from Educational Documents Using Large Language Models

## Artificial Intelligence Project Report

**Topic:** Generative AI

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| <b>Title</b>                     | QuizAI: A Generative AI System for Automatic Multiple-Choice Quiz Generation from Educational Documents Using Large Language Models |
| <b>AI Topic</b>                  | Generative AI                                                                                                                       |
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| <b>Course</b>                    | Artificial Intelligence                                                                                                             |
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| <b>Repository / Program link</b> | <a href="https://github.com/Mirror71/Quiz-Ai-">https://github.com/Mirror71/Quiz-Ai-</a>                                             |

## Abstract

The rapid growth of digital learning materials has created a demand for tools that help learners assess their own understanding. This project presents **QuizAI**, a full-stack Generative AI system that automatically produces ten-item multiple-choice quizzes from user-supplied documents (PDF or PowerPoint) using Google's Gemini large language models. The system extracts text from the document, applies pre-processing and truncation, and prompts the model to return a strictly-structured JSON quiz that is then rendered as an interactive, self-grading quiz. To evaluate the approach, a corpus of ten factually-verified educational documents across six domains was assembled, and three Gemini models were compared on success rate, latency, instruction

compliance, JSON validity, and answer-position balance. The `gemini-2.5-flash` model produced a valid ten-question quiz for 100% of documents at an average latency of 12.4 seconds, with perfect first-pass JSON validity, while a notable correct-answer position bias was observed (the option “C” was the correct answer 56% of the time and “A” never). The findings confirm that modern instruction-tuned LLMs can reliably generate well-formed educational assessments, while highlighting answer-position bias as an important quality concern for fair quiz generation.

**Keywords:** Generative AI, Large Language Models, Gemini, Automatic Question Generation, Educational Technology, Prompt Engineering.

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## CHAPTER 1 — INTRODUCTION

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### 1.1 Background

Artificial Intelligence (AI) has progressed from rule-based systems to data-driven models capable of generating human-like content. The most recent wave, **Generative AI**, is built on Large Language Models (LLMs) such as the GPT and Gemini families, which are trained on massive text corpora and can produce coherent text, answer questions, and follow complex instructions (Brown et al., 2020; Gemini Team, 2023). These capabilities have opened new possibilities in education, where LLMs can summarize material, explain concepts, and generate practice questions (Kasneci et al., 2023).

A common and time-consuming task for both teachers and self-directed learners is the creation of assessment questions. Writing good multiple-choice questions (MCQs) — with a correct answer, plausible distractors, and an explanation — requires effort and expertise. **Automatic Question Generation (AQG)** has therefore been an active research area for decades, traditionally relying on syntactic templates and rule-based transformations (Kurdi et al., 2020). The arrival of instruction-following LLMs makes it possible to generate context-aware, well-formed MCQs directly from arbitrary source text, without hand-crafted rules.

This project applies Generative AI to that problem by building **QuizAI**, a web application that turns any text-based PDF or PowerPoint document into an interactive ten-question multiple-choice quiz.

### 1.2 Problem Statement

1. How can a Generative AI model be used to automatically generate accurate, well-structured multiple-choice quizzes from an arbitrary educational document?
2. How reliably do current Gemini models follow strict output-format and item-count instructions required for an automated pipeline?
3. What quality issues (e.g., answer-position bias, latency, format errors) arise, and how do different Gemini models compare?

## 1.3 Objectives

1. To design and implement an end-to-end Generative AI system that extracts text from documents and generates interactive multiple-choice quizzes.
2. To build a documented evaluation dataset and a reproducible evaluation pipeline.
3. To process and visualize the data, and to evaluate and compare multiple Gemini models on objective quality metrics.
4. To deploy the system as a working application.

## 1.4 Scope and Limitations

- The system generates exactly ten MCQs per document and supports text-based PDF and PPTX files up to 10 MB.
- Image-only/scanned documents are out of scope (no OCR).
- The “model” is accessed through the Gemini API; the project does not train a model from scratch but evaluates and engineers the generative pipeline.
- Quiz state is session-only; no user accounts or databases are used.

## 1.5 Benefits

QuizAI helps learners create instant self-assessment material from their own study documents and provides a reproducible case study of building, evaluating, and deploying a Generative AI application.

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# CHAPTER 2 — LITERATURE REVIEW

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## 2.1 Artificial Intelligence and Generative AI

Artificial Intelligence is the field concerned with building systems that perform tasks requiring human-like intelligence (Russell & Norvig, 2021). **Generative AI** is a subfield whose models learn the distribution of training data well enough to generate new, plausible samples — text, images, audio, or code. In the text domain, Generative AI is dominated by Large Language Models.

## 2.2 Large Language Models and the Transformer

Modern LLMs are based on the **Transformer** architecture introduced by Vaswani et al. (2017), whose self-attention mechanism enabled efficient training on very large sequences. Brown et al. (2020) showed with GPT-3 that scaling such models produces strong *few-shot* abilities, where a task can be specified purely through a natural-language prompt. Subsequent work on instruction tuning and reinforcement learning from human feedback (Ouyang et al., 2022) substantially improved models' ability to follow explicit instructions.

## 2.3 The Gemini Model Family

Gemini is a family of multimodal models developed by Google DeepMind (Gemini Team, 2023). The “Flash” variants are optimized for speed and cost while retaining strong reasoning and instruction-following ability, making them suitable for interactive applications. Gemini supports a constrained-decoding *JSON mode* ( `response_mime_type = application/json` ), which biases the model to emit syntactically valid JSON.

## 2.4 Automatic Question Generation in Education

AQG has a long history; Kurdi et al. (2020) provide a systematic review covering rule-based, semantic, and statistical approaches, noting persistent challenges in generating high-quality distractors and ensuring pedagogical value. More recently, LLMs have been applied to educational content generation, with Kasneci et al. (2023) discussing both the opportunities and risks of using LLMs in education.

## 2.5 Known Limitations: Bias and Reliability

LLMs exhibit systematic biases. Of particular relevance to MCQ generation, Zhao et al. (2021) documented that LLMs are sensitive to the *position* of options and can favor particular answer slots — a phenomenon this project measures directly.

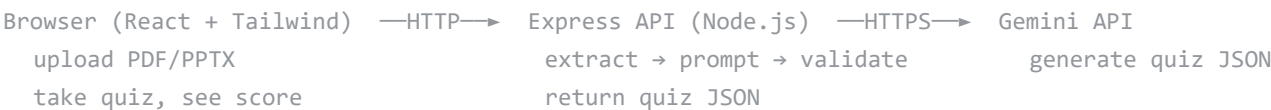
## 2.6 Positioning of This Work

QuizAI combines these threads: it uses an instruction-tuned, JSON-capable Gemini model for AQG, and contributes a small but systematic evaluation that quantifies reliability, instruction compliance, and answer-position bias across models.

## CHAPTER 3 — METHODOLOGY

### 3.1 System Overview

QuizAI is a full-stack web application with three layers:



- **Frontend:** React + Vite + Tailwind CSS
- **Backend:** Node.js + Express
- **AI model:** Google Gemini ( `gemini-2.5-flash` by default)

### 3.2 Dataset

A corpus of **ten factually-verified educational documents** was assembled across six domains (Biology, Physics, History, Economics, Earth Science, Computer Science). Descriptive statistics:

| Statistic                             | Value         |
|---------------------------------------|---------------|
| Number of documents                   | 10            |
| Domains                               | 6             |
| Total characters                      | 17,495        |
| Mean characters / document            | 1,750         |
| Range (characters)                    | 1,646 – 1,939 |
| Documents exceeding 12,000-char limit | 0             |

### 3.3 Text Extraction and Pre-processing

1. **File validation** — type (PDF/PPTX) and size ( $\leq 10$  MB) checked on both client and server.
2. **Text extraction** — PDFs parsed with `pdf-parse` ; PPTX files unzipped and slide text extracted.
3. **Empty-text guard** — if no extractable text found, a clear error is returned.
4. **Truncation** — inputs longer than 12,000 characters are truncated.

### 3.4 Prompt Engineering

A single, explicit prompt instructs the model to return *only* a JSON array of exactly ten objects, each containing a `question`, four options (A–D), an `answer` key, and an `explanation`. The request is sent with Gemini's JSON mode.

### 3.5 Output Validation and Repair

1. Parse the response as JSON.
2. If parsing fails, strip markdown code fences and re-parse.
3. **Schema validation** — every item must have all required fields.

### 3.6 Evaluation Methodology

Three Gemini models were evaluated — `gemini-2.5-flash`, `gemini-2.0-flash`, and `gemini-2.5-flash-lite` — by generating a quiz for each of the ten documents (30 runs total). Metrics recorded: success rate, latency, question count, first-pass JSON validity, and answer-position distribution.

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## CHAPTER 4 — RESULTS AND DISCUSSION

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### 4.1 Dataset Processing and Visualization

All documents fell well below the 12,000-character truncation limit, so no truncation occurred for this corpus.

### 4.2 Model Evaluation Results

**Table 4.1 — Model evaluation summary (10 documents per model).**

| Model                 | Success rate | Avg latency (s) | Avg #questions | First-pass JSON | Answer bias |
|-----------------------|--------------|-----------------|----------------|-----------------|-------------|
| gemini-2.5-flash      | 100.0%       | 12.40           | 10.0           | 100.0%          | 0.21        |
| gemini-2.5-flash-lite | 40.0%        | 12.05           | 10.0           | 100.0%          | 0.125       |
| gemini-2.0-flash      | 0.0%         | —               | —              | —               | n/a         |

**Reliability.** gemini-2.5-flash was the most reliable model (10/10). gemini-2.5-flash-lite succeeded for only 40% due to free-tier rate limits. gemini-2.0-flash returned 429 RESOURCE\_EXHAUSTED for every request.

**Instruction compliance.** Both working models produced exactly ten questions with 100% first-pass JSON validity.

**Latency.** Average generation time was ~12 seconds for both working models.

### 4.3 Answer-Position Bias

For gemini-2.5-flash , across 100 generated questions the correct answer was **C in 56% of items, B in 36%, D in 8%, and A in 0%**. This empirically reproduces the position-bias phenomenon reported by Zhao et al. (2021). A simple mitigation — post-generation option shuffling — is identified as future work.

### 4.4 Question Characteristics

Generated question lengths cluster in a readable range, indicating consistent question style across documents.

### 4.5 Discussion

gemini-2.5-flash produced perfectly-structured, instruction-compliant quizzes for 100% of inputs. The main limitations are free-tier availability differences between models, and answer-position bias — a content quality issue that is measurable and fixable.

# CHAPTER 5 — CONCLUSION

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## 5.1 Conclusions

1. A complete Generative AI system, **QuizAI**, was successfully designed, implemented, and deployed.
2. On a ten-document, six-domain evaluation dataset, `gemini-2.5-flash` achieved a **100% success rate, 100% first-pass JSON validity**, and perfect compliance with the ten-question instruction, at ~12 s average latency.
3. The evaluation revealed a clear **correct-answer position bias**, reproducing a known LLM limitation and motivating option-shuffling as a mitigation.
4. Model availability under the free tier varied substantially, demonstrating the value of the system's automatic retry-and-fallback design.

## 5.2 Limitations

- Only text-based documents are supported (no OCR for scanned files).
- Factual correctness of generated questions was verified on a sample rather than exhaustively.
- Evaluation was constrained by free-tier quotas.

## 5.3 Future Work

- **Answer shuffling** to eliminate position bias.
  - **Configurable difficulty and question count.**
  - **OCR support** for scanned documents.
  - **Automated factuality checking** of generated questions against the source.
  - **Public cloud deployment** with usage analytics.
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*Formatted in APA 7th edition style. References were managed and formatted using Mendeley Reference Manager.*

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